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A novel implementation method of GZZ-based constitutive model into FLAC3D

Yingming Xiao^{a,b}, Manchao He^a, Yafei Qiao^{a,c,*}, Wuqiang Cai^a, Yujing Jiang^b, Hehua Zhu^a^a Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China^b School of Engineering, Nagasaki University, Nagasaki 8528521, Japan^c Key Laboratory of Geotechnical and Underground Engineering of Ministry of Education, Tongji University, Shanghai 200092, China

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ABSTRACT

The Generalized Zhang and Zhu (GZZ) strength criterion is one of the three-dimensional criteria suggested by the International Society for Rock Mechanics and Rock Engineering (ISRM), and it has been widely used and validated. However, the shape of GZZ criterion is non-convex and non-smooth on the π -plane, limiting its implementation into the numerical software. To overcome this limitation, many modified GZZ criteria were proposed to improve the smoothness. Alternatively, a novel approximate numerical implementation method is performed in the mean stress-shear stress-Lode angle space ($\sigma_m - \sqrt{J_2} - \theta_\sigma$) in this paper. It avoids the restriction of yield surface/criterion convexity and smoothness in the principal stress space. Then, a new GZZ-based constitutive model is proposed and implemented into FLAC3D. Moreover, the linearization of the nonlinear GZZ criterion according to the current stress level is used, and this can avoid iteration and greatly improve the computational efficiency. The proposed model is subsequently validated via experimental data, analytical solutions and field data. All results reveal that the proposed model can reproduce various mechanical behaviors of rocks, including nonlinear strength characteristics, dilation behavior and strain-softening behavior. Furthermore, the model can effectively capture the stress redistribution and deformation of rocks induced by tunneling. Therefore, the proposed implementation method of non-convex yield surface/criterion is feasible and able to achieve sufficient calculation accuracy, which provides another possible implementation choice for other yield surfaces/criteria.

1. Introduction

With the exhaustion of shallow resources, human activities have gradually progressed to greater depths (He and Qian, 2010; Zhu et al., 2019; Zhang et al., 2023). In deeply buried underground engineering, most of the current designs are based on the empirical method of two-dimensional stress states (Xiao et al., 2022; Guan et al., 2007), which may lead to severe disasters, such as large deformation (Li et al., 2023) and rockburst (Cheng et al., 2023; He et al., 2022). Meanwhile, the effect of intermediate principal stresses on the mechanical behavior of the surrounding rock has been highlighted (Kong et al., 2021; Zhang et al., 2023; Gu et al., 2023). The influence of the three-dimensional stress state on the deformation and plastic zone of the surrounding rock cannot be neglected (Xu and Xia, 2021; Cai et al., 2023). Therefore, accurately reproducing the mechanical response of deep rock under a three-dimensional stress state is highly important for construction and support design.

To capture the mechanical behavior of rock, the constitutive model based on the two-dimensional strength criterion is widely adopted (Cundall et al., 2003; Ren et al., 2019; Song and Rodríguez-Dono, 2021; Qiao et al., 2023). It assumes that intermediate principal stress does not affect strength and plastic deformation, such as the Hoek-Brown-based and Mohr-Coulomb-based constitutive models in FLAC3D (Xiao et al., 2023; Wang et al., 2022). However, this method has shortcomings in investigating ground reactions, especially the effect of intermediate principal stress on ground reactions. Thus, a series of constitutive models based on the three-dimensional strength criterion is proposed. Assuming that the intermediate principal stress and the minimum principal stress have the same effect on strength and plastic deformation, the Drucker-Prager-based constitutive model was proposed (Itasca, 2005). Feng et al. (2021) proposed a three-dimensional constitutive model, which can capture the 3D stress-induced anisotropy. Considering the nonlinearity of strength, based on the Hoek-Brown criterion and ideas of generalized nonlinear unified strength theory (GNUST), the

* Corresponding author at: Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China.

E-mail address: yafei.qiao@tongji.edu.cn (Y. Qiao).

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GNUST-based constitutive model was proposed (Wang et al., 2023). Nevertheless, the GZZ criterion has a high fitting accuracy for true triaxial strength, which has been validated by many experimental data. This criterion has been recommended by the International Society for Rock Mechanics and Rock Engineering (ISRM) in “Three-dimensional failure criteria based on the Hoek-Brown criterion”.

The GZZ criterion has the shape of the Mogi–Coulomb criterion on the π -plane. Its shape is non-smooth and non-convex (Zhu et al., 2013). The non-smoothness of the GZZ criterion is located in the triaxial compression (TC) state and triaxial extension (TE) state; the non-convexity is located in the TE state (Zhang et al., 2013). Between the TC and TE states, the GZZ criterion is a smooth convex curve. In general, the form of the strength criterion could be adopted as a yield surface equation. However, during numerical calculations, non-smoothness and non-convexness of the yield surface may lead to several difficulties. For instance, the yield function is nondifferentiable (Sorensen et al., 2015) at the non-smoothness point, and the solution is nonunique around the non-convexity point. All these issues can lead to non-convergence of the iteration or simulation.

To implement the non-convexity and non-smooth yield function into the numerical software, two methods are often adopted. The first method is smoothing (Wang, 1992) and convexifying the yield function (Zhu et al., 2017) in the principal stress space. Zhang et al. (2013) modified the Lode dependence of the GZZ criterion by employing three Lode dependence functions with smoothness and convexity characteristics, including the elliptical dependence (William and Warnke, 1975), the hyperbolic dependence (Yu, 1990) and the spatial mobilized (Mat-suoka and Nakai, 1974). Zhu et al. (2017) evaluated the characteristics of three types of Lode dependence functions via numerical implementation. Then, a constitutive model with hyperbolic dependence is established and implemented in the 3D FE code GeoFBA3D. To obtain the peak strength directly at a certain intermediate and minimum principal stress, Chen et al. (2021) proposed a newly modified GZZ criterion, which has an explicit expression with a smooth and convex shape on the π -plane. The newly modified GZZ criterion-based constitutive model is implemented in FLAC3D. Cai et al. (2021) developed a smoothed GZZ criterion for high-stressed deep rock mass. Alternatively, a new stress space (e.g., octahedral shear stress and mean normal stress space) can be adopted to avoid the non-convexity and non-smoothness issues of specific yield functions. For instance, Feng et al. (2019) implemented a Mogi-based constitutive model into FLAC3D by adopting octahedral shear stress and an effective mean normal stress space. Thus, the latter method can also offer a possible way for the implementation of GZZ criterion.

Therefore, an alternative numerical implementation method for the GZZ criterion was proposed, and this process was performed in the mean stress-shear stress-Lode angle ($\sigma_m - \sqrt{J_2} - \theta_\sigma$) space. In addition, a new explicit calculation format for the GZZ-based constitutive model was formulated, and the nonlinear failure surface of GZZ was continuously approximated by the linear tangent at the current stress level. The work in this paper is distinct based on the following two characteristics:

- An alternative numerical implementation framework of the GZZ-based constitutive model, of which the yield surface is non-convexity and non-smoothness, in the stress invariant space ($\sigma_m - \sqrt{J_2} - \theta_\sigma$) is proposed;
- The GZZ criterion is approximately linearized in the $\sigma_m - \sqrt{J_2}$ space to avoid iterations, then the GZZ-based constitutive model is formulated and implemented into FLAC3D.

In the following sections, the GZZ criterion is first briefly reviewed. After that, the numerical implementation method for the GZZ criterion is presented. A GZZ-based model is then proposed and implemented into FLAC3D. Finally, the proposed constitutive model is verified from laboratory tests, theoretical analysis, and case study.

2. A brief review of the GZZ criterion

The GZZ criterion is proposed based on the Hoek-Brown criterion and the empirical formula for rock strength suggested by Mogi for predicting true triaxial strength. The original Hoek-Brown criterion (Hoek and Brown, 1980; Hoek and Brown, 1997) is used to reproduce the strength of intact rocks, reflecting the nonlinear increase in peak strength with increasing confining pressure. It can be expressed as follows:

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_i \frac{\sigma_3}{\sigma_c} + 1 \right)^{0.5} \quad (1)$$

where σ_1 and σ_3 are the major and minor principal stresses, respectively; m_i is the material constant of intact rock; and σ_c denotes the uniaxial compressive strength.

The intact rock and the rock mass clearly differ, especially in strength. To capture the strength of the rock mass, Hoek et al. (2002), Hoek and Brown (2019) extended the Hoek-Brown criterion to the generalized Hoek-Brown criterion as follows:

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a \quad (2)$$

$$m_b = m_i e^{(GSI-100)/(28-14D)} \quad (3)$$

$$s = e^{(GSI-100)/(9-3D)} \quad (4)$$

$$a = \frac{1}{2} + \frac{1}{6} \left(e^{\left(\frac{-GSI}{15} \right)} - e^{\left(\frac{-30}{15} \right)} \right) \quad (5)$$

where m_b is the reduced value of m_i , which considers the decrease in rock mass strength relative to intact rock; s and a are material constants of the rock mass; GSI is the geological strength index (GSI); and D is the disturbance factor representing the level of blast damage and stress relaxation to the rock mass.

The generalized Hoek-Brown criterion is widely used for two reasons: (i) the good capacity to estimate the strength of the rock mass and (ii) the ability to obtain mechanical parameters directly from the rock mass coupled with the uniaxial compressive strength of the intact rock (Eberhardt, 2012).

Mogi-Coulomb criterion (Mogi, 1967, 1971, 2006) can well reproduce the experimental data considering the effect of intermediate principal stress on the strength of rock masses. Combining the advantages of the Hoek-Brown criterion for two-dimensional strength estimation and the Mogi-Coulomb criterion (Mogi, 2006) for three-dimensional strength estimation, Zhang and Zhu (2007) proposed a three-dimensional Hoek-Brown criterion, which can be expressed as follows:

$$\frac{9}{2\sigma_c} \tau_{oct}^2 + \frac{3}{2\sqrt{2}} m_b \tau_{oct} - m_b \sigma_{m,2} = s \sigma_c \quad (6)$$

$$\tau_{oct} = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} \quad (7)$$

$$\sigma_{m,2} = \frac{\sigma_1 + \sigma_3}{2}$$

where τ_{oct} and $\sigma_{m,2}$ are the octahedral shear stress and effective mean normal stress, respectively; and σ_2 is the intermediate principal stress.

For jointed rock masses, Zhang (2008) proposed a generalized three-dimensional Hoek-Brown strength criterion (GZZ criterion), which can be expressed as follows:

$$\frac{1}{\sigma_c \left(\frac{1}{a} - 1 \right)} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right)^{1/a} + \frac{m_b}{2} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right) - m_b \sigma_{m,2} = s \sigma_c \quad (8)$$

where the mechanical parameters (m_b , s , a) are the same as those of the

Hoek-Brown criterion.

Fig. 1 shows the shape of the GZZ criterion in the principal stress ($\sigma_1 - \sigma_2 - \sigma_3$) space. The GZZ criterion yields a non-smooth concave cone (Fig. 1a). Fig. 1b compares the shapes of the GZZ criterion and the GZZ-related criterion on the π -plane. GZZ-Zhang, GZZ-Chen and GZZ-Cai represent the modified GZZ criterion proposed by Zhang et al. (2013), Chen et al. (2021) and Cai et al. (2021), respectively. They are almost coincide with each other. Nonetheless, the GZZ criterion has a simple mathematical form and clear physical meaning.

3. Numerical implementation method of GZZ criterion

3.1. GZZ criterion expression in the $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space

To address the non-convexity and non-smoothness of the GZZ criterion in $\sigma_1 - \sigma_2 - \sigma_3$ space, the GZZ criterion is expressed in terms of σ_m , $\sqrt{J_2}$, and θ_σ as follows:

$$\frac{1}{\left(\frac{1}{a}-1\right)}(\sqrt{3J_2})^{1/a} + \left(\frac{\sqrt{3}}{2} - \frac{\sin\theta_\sigma}{\sqrt{3}}\right)m_b\sqrt{J_2} - m_b\sigma_m - s\sigma_c = 0 \quad (9)$$

in which

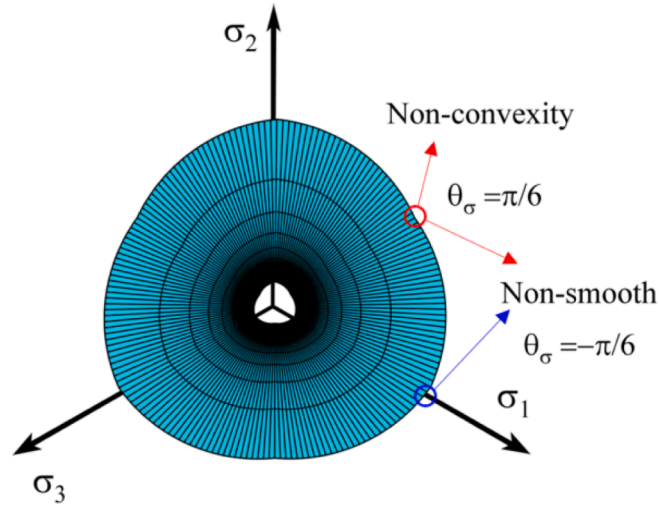
$$\sigma_m = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3} \quad (10)$$

$$J_2 = \frac{1}{6}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] \quad (11)$$

$$\theta_\sigma = \tan^{-1}\left[\frac{\sigma_1 + \sigma_3 - 2\sigma_2}{\sqrt{3}(\sigma_1 - \sigma_3)}\right] \quad (12)$$

Fig. 2 shows the shape of the GZZ criterion in the $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space. The GZZ criterion is smoothness and non-convexness in $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space, as shown in Fig. 2a. Moreover, intercepting the GZZ criterion in a certain plane perpendicular to the Lode angle axis, as shown in Fig. 2b, the shape of the GZZ criterion in the stress space of $\sigma_m - \sqrt{J_2}$ is convex

(a)



(b)

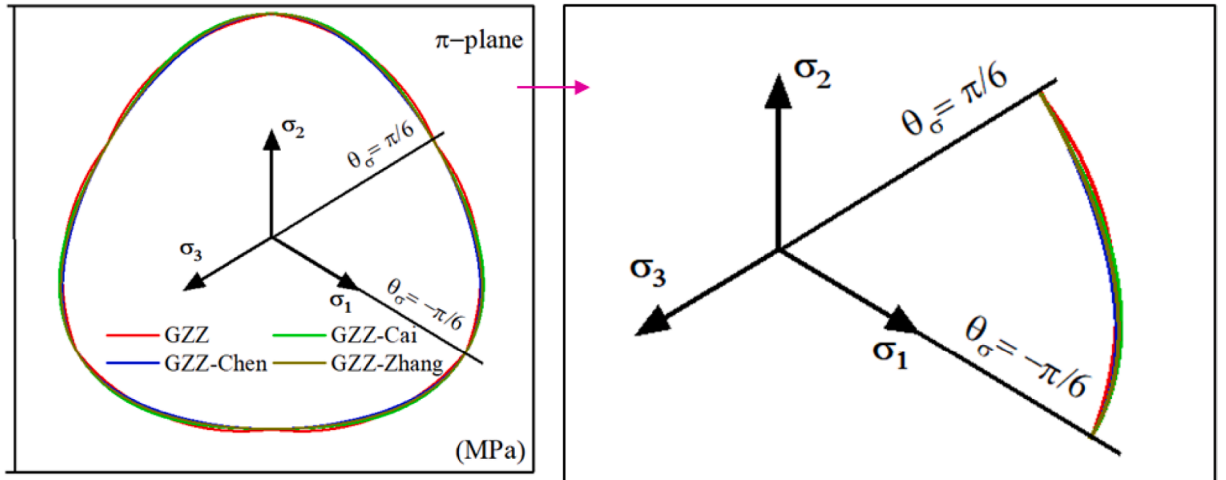


Fig. 1. Envelope of the GZZ and related criterion in the principal stress space: (a) the spatial envelope of the GZZ criterion and (b) the envelope of the GZZ-related criterion on the π -plane.

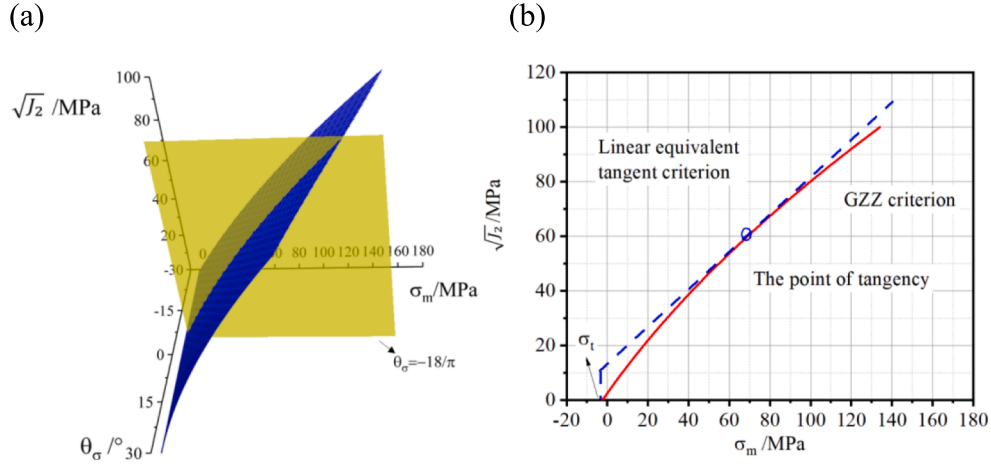


Fig. 2. Envelope of the GZZ criterion: (a) the shape of the GZZ in the stress-invariant space $\sigma_m - \sqrt{J_2} - \theta_\sigma$ and (b) the linear equivalent tangential criterion of the GZZ criterion in $\sigma_m - \sqrt{J_2}$ plane at $\theta_\sigma = -\frac{18}{\pi}$.

and smooth.

3.2. Stress return in $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space

According to elastoplastic theory (Zheng et al., 2002; Clausen and Damkilde, 2008), the total strain increment $\Delta \epsilon$ can be divided into an elastic strain increment $\Delta \epsilon^e$ and a plastic strain increment $\Delta \epsilon^p$, which can be expressed as follows:

$$\Delta \epsilon = \Delta \epsilon^e + \Delta \epsilon^p \quad (13)$$

The new stress vector σ^N can be calculated as

$$\sigma^N = \sigma^p + \Delta \sigma \quad (14)$$

in which the superscripts N and P represent the information of the current step and the previous step, respectively. $\Delta \sigma$ is the stress increment.

$$\Delta \sigma = D^e (\Delta \epsilon - \Delta \epsilon^p) = \Delta \sigma^e - \Delta \sigma^p \quad (15)$$

in which $\Delta \sigma^e$ and $\Delta \sigma^p$ are the stress increments decomposed to the elastic part and plastic corrector stresses, respectively, and D^e is the elasticity matrix.

Then the calculation of the stress increment will be divided into two steps, the calculation of the elastic guess stress σ^J and the calculation of the new stress vector σ^N . They can be calculated as follows,

$$\sigma^J = \sigma^p + \Delta \sigma^e \quad (16)$$

$$\sigma^N = \sigma^J - \Delta \sigma^p \quad (17)$$

In the $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space, Eq. (14) can be rewritten as follows:

$$q^N = q^p + G \Delta \gamma^e \quad (18)$$

$$\sigma_m^N = \sigma_m^p + K \Delta \epsilon_m^e \quad (19)$$

in which K and G are the bulk modulus and shear modulus, respectively. The vector q represents the magnitude $\sqrt{J_2}$ and direction θ_σ of shear stress. The vector $\Delta \gamma^e$ represents the magnitude and direction of elastic shear strain.

According to Eq. (15), the plastic corrector stresses can be calculated as follows in the $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space:

$$\Delta \sigma_m^p = K \Delta \epsilon_m^p \quad (20)$$

$$\Delta q^p = G \Delta \gamma^p \quad (21)$$

$$\Delta \gamma^p = \Delta \gamma_{\sqrt{J_2}}^p + \Delta \gamma_\theta^p \quad (22)$$

in which $\Delta \sigma_m^p$ and Δq^p are the plastic stress increments at the hydrostatic pressure and plastic stress deviation, respectively. Δq^p comprises $\Delta \sqrt{J_2}^p$ and $\Delta \theta_\sigma^p$, the shear stress increment and the Lode angle increment, respectively. $\Delta \epsilon_m^p$ and $\Delta \gamma^p$ are the plastic volumetric strain increment and plastic shear strain increment, respectively. $\Delta \epsilon_m^p$ is caused by the hydrostatic pressure σ_m , and $\Delta \gamma^p$ is caused by shear stress. $\Delta \gamma_{\sqrt{J_2}}^p$ and $\Delta \gamma_\theta^p$ are the two components of plastic shear strain, as shown in Fig. 3.

$\Delta \theta_\sigma^p$ is caused by $\Delta \gamma_\theta^p$, and $\Delta \gamma_\theta^p$ is often very small (Zheng et al., 2002). In this work, an approximation algorithm is used to consider $\Delta \theta_\sigma^p$; the Lode angle at the current step θ_σ^N is assumed as follows:

$$\theta_\sigma^N = k \theta_\sigma^p + (1 - k) \theta_\sigma^J \quad (23)$$

in which k is the weighting factor in the range of 0–1; and θ_σ^p , θ_σ^J and θ_σ^N are the Lode angles of the previous step, elastic guess and current step, respectively. As indicated in Eq. (23), the Lode angle is a certain value at each constitutive calculation regardless of the k values, meaning that the computational stress space is in a certain plane perpendicular to the Lode angle axis. In particular, to simplify the calculation, k can be assumed to be 0. In this case, $\theta_\sigma^N = \theta_\sigma^J$. Once θ_σ^N is determined, the computations of $\Delta \sigma_m^p$ and Δq^p can be simplified to calculation in two-dimensional stress space ($\sigma_m - \sqrt{J_2}$ in Fig. 2). The calculation

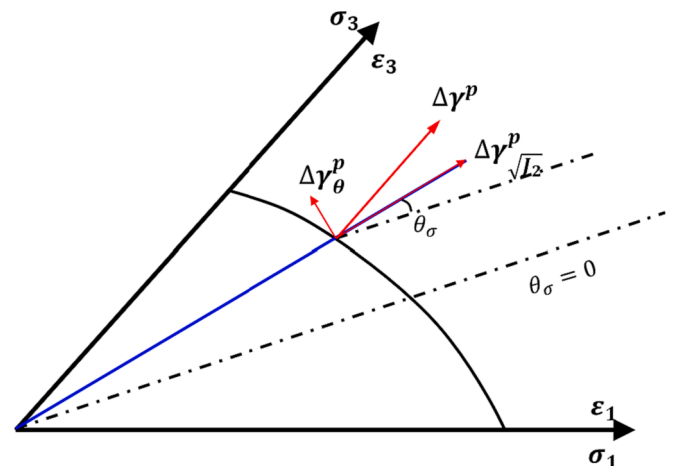


Fig. 3. Geometric relationship of the flow law on the π -plane.

procedure is presented in Section 4.4.

4. The proposed GZZ-based model

The GZZ criterion is adopted as the yield surface equation, which is expressed in the stress space of $\sigma_m - \sqrt{J_2}$. In addition, the non-associated flow rule is employed to capture the dilation behavior. The strain-softening is simulated by combining the linear hardening rule. Then, a novel GZZ-based model is proposed and implemented into FLAC3D. Finally, the model performance is investigated.

4.1. Yield surface equation

The yield surface equation adopts the form of the GZZ criterion. Owing to the θ_σ^N has been determined, then, the yield surface equation can be rewritten as follows,

$$F = \frac{1}{\left(\frac{1}{a}-1\right)} (\sqrt{3J_2})^{1/a} + nm_b \sqrt{J_2} - m_b \sigma_m - s \sigma_c \quad (24)$$

$$n = \left(\frac{\sqrt{3}}{2} - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) \quad (25)$$

The yield surface equation (Eq. (24) in the stress space of $\sigma_m - \sqrt{J_2}$ is nonlinear (Fig. 2b), which may lead to iterative calculations during the stress update of each element. In particular, for FLAC3D software, constitutive computation is required for each element under each step. This will significantly reduce the computational efficiency.

To avoid iterative computations and improve computational efficiency, as shown in Fig. 2b, the nonlinear criterion of GZZ in $\sigma_m - \sqrt{J_2}$ space is continuously approximated by the linear tangent at the current stress level of θ_σ and $\sqrt{J_2}$. The current linear yield surface f is expressed as follows:

$$f = \sqrt{J_2} - q_\phi \sigma_m + k_\phi = 0 \quad (26)$$

$$q_\phi = \frac{\partial F}{\partial \sigma_m} / \frac{\partial F}{\partial \sqrt{J_2}} \quad (27)$$

$$k_\phi = \sqrt{J_2} + \frac{\partial F}{\partial \sigma_m} \sigma_m^{fit} / \frac{\partial F}{\partial \sqrt{J_2}} \quad (28)$$

in which

$$\frac{\partial F}{\partial \sigma_m} = -m_b \quad (29)$$

$$\frac{\partial F}{\partial \sqrt{J_2}} = \frac{1}{a \sigma_c} (\sqrt{3J_2})^{\frac{1}{a}-1} + \left(\frac{\sqrt{3}}{2} - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \quad (30)$$

$$\sigma_m^{fit} = \left[\frac{1}{\left(\frac{1}{a}-1\right)} (\sqrt{3J_2})^{\frac{1}{a}} + \left(\frac{\sqrt{3}}{2} - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \sqrt{J_2} - s \sigma_c \right] / m_b \quad (31)$$

4.2. Non-associated flow rule

The plastic potential function takes a similar form to the strength criterion. A dilation coefficient η is introduced to characterize the dilation of the rock. The plastic potential function g is defined as follows:

$$g = \frac{1}{\left(\frac{1}{a}-1\right)} (\sqrt{3J_2})^{1/a} + \left(\frac{\sqrt{3}}{2} - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \sqrt{J_2} - \eta m_b \sigma_m - s \sigma_c = 0 \quad (32)$$

in which η is the dilation coefficient. η takes values from 0 to 1. When $\eta = 1$, the flow rule is associated; otherwise, the flow rule is non-associated. In particular, when $\eta = 0$, the volume remains constant.

The plastic strain increment can be calculated as follows:

$$\Delta \varepsilon_m^p = \Delta \lambda \frac{\partial g}{\partial \sigma_m} \quad (33)$$

$$\Delta \gamma^p = \sqrt{\Delta \gamma_{\sqrt{J_2}}^p{}^2 + \Delta \gamma_\theta^p{}^2} = \Delta \lambda \sqrt{\left(\frac{\partial g}{\partial \sqrt{J_2}} \right)^2 + \left(\frac{1}{\sqrt{J_2}} \frac{\partial g}{\partial \theta_\sigma} \right)^2} \quad (34)$$

in which $\Delta \lambda$ is the plastic multiplier.

4.3. Hardening rule

The hardening equation is often a relationship between mechanical parameters (m_b , s and a) and plastic internal variable γ_{inter}^p . The plastic internal variable γ_{inter}^p is often employed to record the plastic history, and the following form is adopted in the proposed model:

$$\gamma_{inter}^p = \sqrt{\frac{(\varepsilon_1^p - \varepsilon_2^p)^2 + (\varepsilon_2^p - \varepsilon_3^p)^2 + (\varepsilon_3^p - \varepsilon_1^p)^2}{2}} = \frac{1}{\sqrt{3}} \gamma^p \quad (35)$$

in which ε_1^p , ε_2^p and ε_3^p are the maximum, intermediate and minimum principal plastic strains, respectively.

The 'table function' in FLAC3D can be used to realize the hardening/softening behavior of a rock mass. Three specific tables can be provided to control the evolution of m_b , s and a . The specific table consists of a series of point pairs. Each point pair is composed of the plastic internal variable γ_{inter}^p and mechanical parameters. According to the accumulated plastic shear strain and the provided tables, the material parameters are updated. Taking parameter m_b as an example, the point pair is in the form of (γ_{inter}^p, m_b) . Fig. 4 shows the evolution of m_b with γ_{inter}^p . The specific table of m_b is ((0,17) (0.001,13) (0.002, 10.3) (0.003,10)). The neighboring pairs of points are linearly connected.

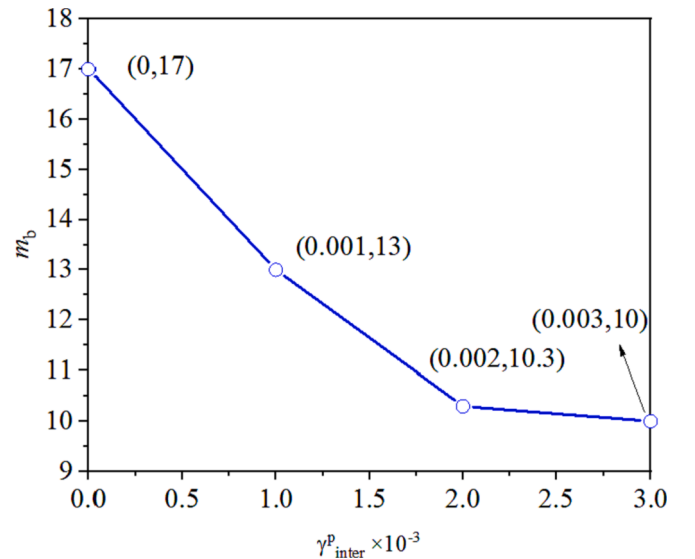


Fig. 4. Evolution of the mechanical parameters m_b with respect to the plastic internal variables.

4.4. Stress update for the GZZ-based model

All calculated elastic guess stresses are then converted to the $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space, which is substituted into the linear equivalent tangential yield surface equation f (Eq. (26)). When $f < 0$, the stress state is in the elastic state, and the elastic guess stresses are equal to the new stress state (σ^N).

If $f \geq 0$, the plastic corrector stress would be calculated. By substituting Eqs (33)–(35) into Eqs. (20) and (21), the plastic corrector stress can be obtained as follows:

$$\Delta\sigma_m^p = \Delta\lambda K \frac{\partial g}{\partial \sigma_m} \quad (36)$$

$$\Delta q^p = \Delta\lambda G \sqrt{\left(\frac{\partial g}{\partial \sqrt{J_2}}\right)^2 + \left(\frac{1}{\sqrt{J_2}} \frac{\partial g}{\partial \theta_\sigma}\right)^2} \quad (37)$$

Considering that $\Delta\gamma_\theta^p$ is very small, $k = 0$ in Eq. (23) is assumed to simplify the calculation. In other words, in each calculation, the non-circular plastic potential function is approximated by a circular plastic potential function at the current stress level.

Then the new stress state can be obtained as follows:

$$\sigma_m^N = \sigma_m^I - \Delta\lambda K \frac{\partial g}{\partial \sigma_m} \quad (38)$$

$$\sqrt{J_2}^N = \sqrt{J_2}^I - \Delta\lambda G \frac{\partial g}{\partial \sqrt{J_2}} \quad (39)$$

$$\theta_\sigma^N = \theta_\sigma^I \quad (40)$$

Considering the linearized equivalent tangent criterion of the GZZ criterion, the plastic multipliers can be calculated as follows:

$$\Delta\lambda = \frac{f(\sigma_m^I, \sqrt{J_2}^I, \theta_\sigma^I)}{G + \eta K q_\phi} \quad (41)$$

Substituting Eq. (41) into Eq. (34), the plastic shear strain increment can be calculated. Meanwhile, the plastic shear strain γ^p can be calculated as follows,

$$\gamma_p = \gamma_p^p + \Delta\gamma_p \quad (42)$$

where γ_p^p is the accumulated plastic shear strain of the previous step. Finally, the new incremental stresses are converted to the Cartesian stress state (Itasca, 2005). The calculation flowchart in FLAC3D is summarized in Fig. 5. More details can be found in Appendix A and Appendix B.

4.5. Model performance

The model parameters are first summarized, followed by their calibration method. The mechanical behavior of the Mizuho trachyte was well tested through a series of true triaxial tests and reported in the literature (Mogi, 1971; Chen et al., 2021). Thus, the Mizuho trachyte is adopted to illustrate the model performance, including the dilation

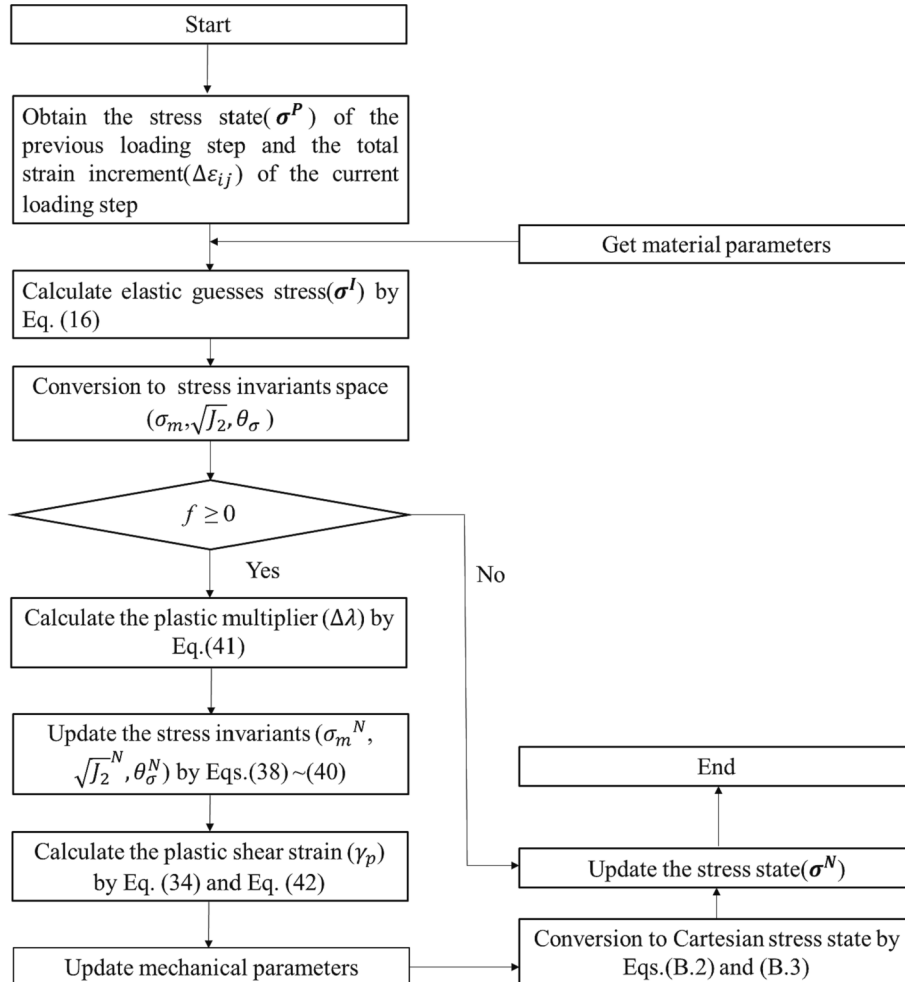


Fig. 5. Calculation flowchart of GZZ constitutive model in FLAC3D.

behavior and strain-softening behavior. The mechanical parameters are presented in Table 1 and their calibration process is presented in Section 5.1.

4.5.1. Model parameters

The proposed GZZ model has a total of 7 parameters and 3 specific tables, including 4 strength-dependent parameters (σ_c , m_b , s and a), 1 dilation-dependent parameter η and 2 elastic parameters (Elastic modulus E and Poisson ratio ν). Strength-dependent parameters affects the peak strength of rock mass. m_b , s and a can be obtained according to m_i , GSI and D of Eqs. (3)–(5). σ_c can be determined by the uniaxial compression test (Hoek and Brown, 2019); the dilation-dependent parameter η can affect the dilation behavior during the plastic stage. It can be calibrated by back-fitting the stress–strain curve; elastic parameters can be determined by the classical method; Adopting the ‘table function’ of FLAC3D, 3 specific tables can control the evolution of m_b , s and a .

4.5.2. Description of the dilation behavior

To illustrate the dilation characteristics, a parametric sensitivity analysis of the dilation coefficient η is performed. η are taken as 0.0, 0.5, and 1.0. Fig. 6 demonstrates the axial strain–volumetric strain curves of the Mizuho trachyte at $\sigma_2 = 20$ MPa and $\sigma_3 = 10$ MPa. As η increases, the volumetric strain curve gradually shifts upward, and the volumetric expansion increases. In particular, at $\eta = 0$, the volume expansion of the rock is 0.

4.5.3. Description of the strain-softening behavior

To illustrate the capacity of the proposed constitutive model to describe the strain-softening behavior of rocks, as shown in Fig. 7a, three types of stress–strain curves are simulated. Type I, II and III exhibit elastic–brittle–plastic characteristics, strain-softening characteristics and elastic–perfectly plastic characteristics, respectively. The peak strength parameters are all the same (Table 1). The ‘table function’ is employed when simulating the stress–strain curves of Type I and II. Two parameters (m_b and s) evolve with the plastic internal variable γ_{inter}^p . As shown in Fig. 7b, both the final mechanical parameters (m_b and s) evolved to 6.03 and 0.17, respectively. The difference is that the plastic internal variable corresponding to the final value is 0.037 in Type I, while 0.067 in Type II. Moreover, to describe the nonlinear behavior of the post-peak stage, multiple point pairs can be used to describe the relationship between the plastic internal variables and the three mechanical parameters (m_b , s and a).

5. Model validation

The capacity of the proposed GZZ-based model is validated in this section. First, the proposed method was validated by two sets of strength data and one set of stress–strain curve data obtained via laboratory tests. Second, the stress redistribution induced by the excavation of a circular tunnel is predicted by the proposed GZZ-based model and compared to its semi-analytical solution. Finally, the model is used to analyze the tunneling-induced deformation of one highway tunnel reported in the literature.

Table 1

All the mechanical parameters for the KTB Amphibolite, Westly granite and Mizuho trachyte (modified after Zhang, 2008; Chen et al., 2021).

Rock	σ_c (MPa)	m_b	s	a	E^* (GPa)	ν^*
KTB Amphibolite	165.0	31.0	1.0	0.5	–	–
Westly granite	201.0	35.0	1.0	0.5	–	–
Mizuho trachyte	100	10.55	1.0	0.5	13	0.27

* The elastic modulus E and Poisson ratio ν are only required when calculating the stress–strain curve.

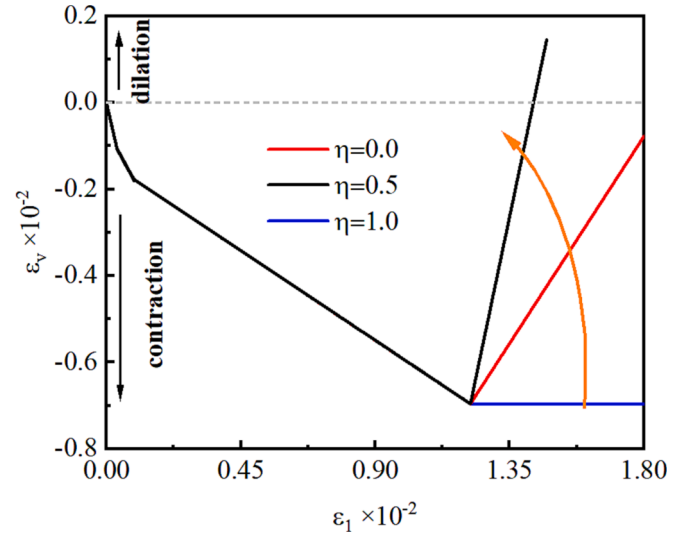


Fig. 6. Effect of the dilation coefficient on the volumetric strain.

5.1. True triaxial compression

Adopting KTB Amphibolite and Westly granite to validate the reproducibility of the proposed GZZ-based constitutive model for peak strength. KTB Amphibolite and Westly granite are both intact rocks, thus their GSIs are 100. The best-fitting values of m_i reported by Zhang and Zhu (2007) are adopted. Then, m_b , s and a can be easily obtained according to Eqs (3)–(5). The model parameters are summarized in Table 1.

As shown in Fig. 8, the peak strengths of KTB Amphibolite and Westly granite in numerical are in good agreement with the test results. In particular, when the stress level of σ_2 is low, the phenomenon of a gradual increase in peak strength as σ_2 increases can be captured.

The true triaxial test result of the Mizuho trachyte (Mogi, 1971; Chen et al., 2021) is also adopted to further validate the correctness of the GZZ constitutive model in terms of stress–strain curves. Fig. 9 shows the axial stress–axial strain curves of the Mizuho trachyte at $\sigma_3 = 35$ MPa with different values of σ_2 . The test results show that the rock exhibits elastic–perfectly plastic behavior. Thus, the GZZ-based model without a ‘table function’ is adopted herein. All the tests were used to calibrate the model parameters, and the parameters are shown in Table 1. Fig. 9 shows the simulated results of the GZZ model, which agree well with the experimental results.

5.2. Stress redistribution induced by the excavation of a circular tunnel

In this section, the stress distribution induced by the excavation of a circular tunnel is simulated and predicted by using the proposed GZZ-based constitutive model. A FLAC3D model is established, as shown in Fig. 10. The model obeys the following assumptions:

- The model radius R is 120 m. The tunnel excavation radius is $r_0 = 8$ m, and a radius of plastic R_p would form under hydrostatic pressure.
- A hydrostatic pressure p_0 is applied at the outer boundary of the rock mass. A uniform support pressure p_i is assumed to act radially on the tunnel wall, and an axial stress of σ_z is applied in the axial direction of the tunnel;
- The mass is assumed to be continuous, homogeneous, isotropic, and initially elastic;
- The proposed GZZ model is applied. The constitutive models of elastic–perfectly plastic (EP) and elastic–brittle–plastic (EBP) are used. The mechanical parameters are summarized in Table 2. GSI is the GSI based on residual strength equivalent calculation. assuming no volume change at the plastic zone, $\eta = 0$;

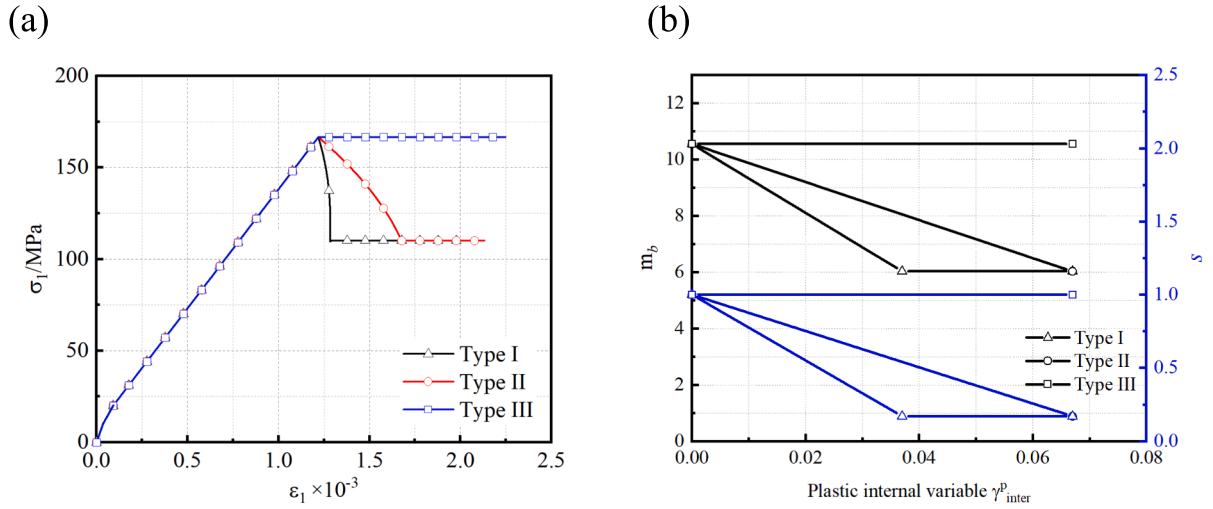


Fig. 7. Model response of the simulated strain-softening behavior: (a) stress–strain curves; (b) the evolution of m_b and s . Types I, II and III represent elastic–brittle–plastic, strain-softening and elastic–perfectly plastic behaviors, respectively.

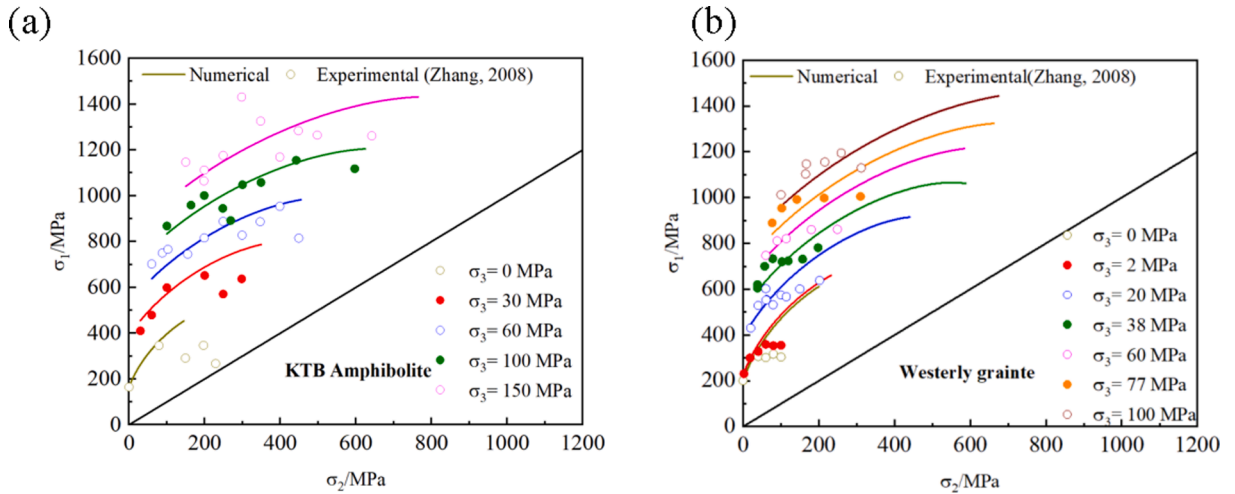


Fig. 8. Comparison of the strengths from the experimental data and numerical results: (a) KTB Amphibolite; and (b) Westerly granite. (Experimental data from Chang, 2001).

The stress distributions simulated by the proposed model are compared with the analytical results reported by Su et al. (2020) in Fig. 11. The numerical results agree well with the analytical results. Adopting the elastic–perfectly plastic constitutive model (Fig. 11a), the radial stress σ_r gradually increases. The tangential stress σ_θ first increases and then decreases, and its maximum value ($\sigma_{\theta max}$) represents the boundaries of the plastic zone. The axial stress σ_z gradually increases in the plastic zone and remains constant in the elastic zone. When adopting the elastic–brittle–plastic constitutive model (Fig. 11b), a significant variation can be observed in the distribution of σ_r , σ_θ and σ_z . Within the plastic zone, σ_r , σ_θ and σ_z increase gradually. At the boundary of the plastic zone, the peak strength increases sharply, causing σ_θ shows a sharp increase; Within the elastic zone, the growth rate of σ_r accelerates, and then σ_θ gradually decreases and σ_z remains almost constant.

5.3. Deformation induced by the excavation of a highway tunnel

A highway tunnel in Guizhou with detailed construction and monitoring records was used to validate the ability of the proposed constitutive model. The buried depth of the tunnel is around 140 m. The tunnel was excavated using the double-step excavation method. The surrounding rock of the tunnel is sandstone, and the uniaxial

compression strength is 13.2 MPa. The GSI and m_i are 47 and 17, respectively (reported by Zhu et al., 2017). The lateral pressure coefficient is 1.0 (Chen et al., 2021).

A 75 m long section of the tunnel is selected to conduct the numerical simulation. A 3D numerical model is built by FLAC3D, as shown in Fig. 12. To eliminate the boundary effect, the width of the model is 75 m. To reproduce the initial in situ stress, a vertical stress of 1.54 MPa is applied to the top of the model to simulate the overburden, and the remaining boundaries are fixed in the normal direction.

There are a total of 115 construction steps, including up-bench excavation, up-bench initial lining, down-bench excavation, down-bench initial lining, backfill, and second lining installation. The length of both bench excavations and initial support is 2 m/step. The length of the backfill and second lining installation is 10 m/step. Table 3 summarizes the material parameters of the support.

The GZZ-based model without the ‘table function’ is adopted to capture the mechanical behavior of the sandstone. Owing to the adopted blast drilling method, the surrounding rock parameters are reduced. The surrounding rock was divided into two parts, disturbed rock mass and undisturbed rock mass. The horizontal and vertical extents of the tunnel at 9 m and 6 m, respectively, are considered disturbed rock masses, as shown in Fig. 12. According to Chen et al. (2021) and Wang et al.

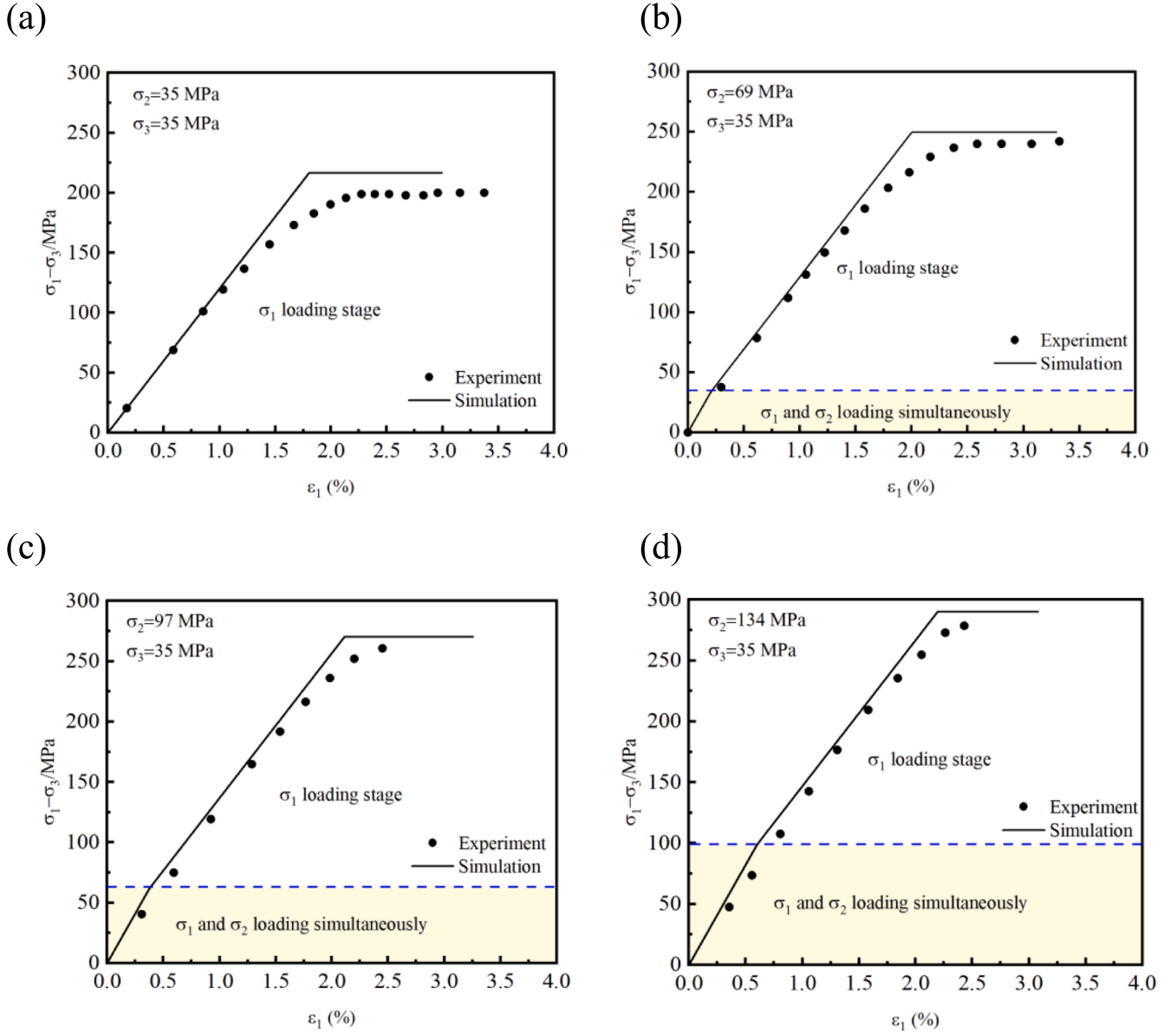


Fig. 9. Comparison of the axial stress-axial strain curves of the Mizuho trachyte from the experimental data and numerical results. (a) $\sigma_2 = 35\text{MPa}$; (b) $\sigma_2 = 69\text{MPa}$; (c) $\sigma_2 = 97\text{MPa}$; and (d) $\sigma_2 = 134\text{MPa}$. (Experimental data from Mogi, 1971; Chen et al., 2021).

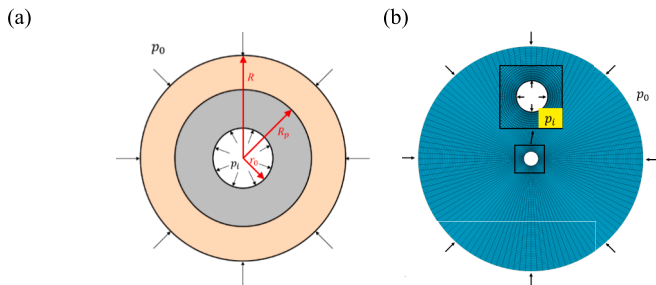


Fig. 10. Circular tunnel subjected to hydrostatic pressure: (a) mechanical model; (b) numerical model.

(2023), the disturbance coefficients D of the GZZ criterion are 0.3 and 0 for disturbed rock and undisturbed rock, respectively. The elastic modulus is determined according to the method proposed by Hoek and Diederichs (2006). Table 4 lists the material parameters of the surrounding rock. Notably, all the basic model parameters used are the

same as those reported in the literature (Zhu et al., 2017; Chen et al., 2021; Wang et al., 2023). This is a benefit for the comparison of different versions of GZZ-based constitutive models.

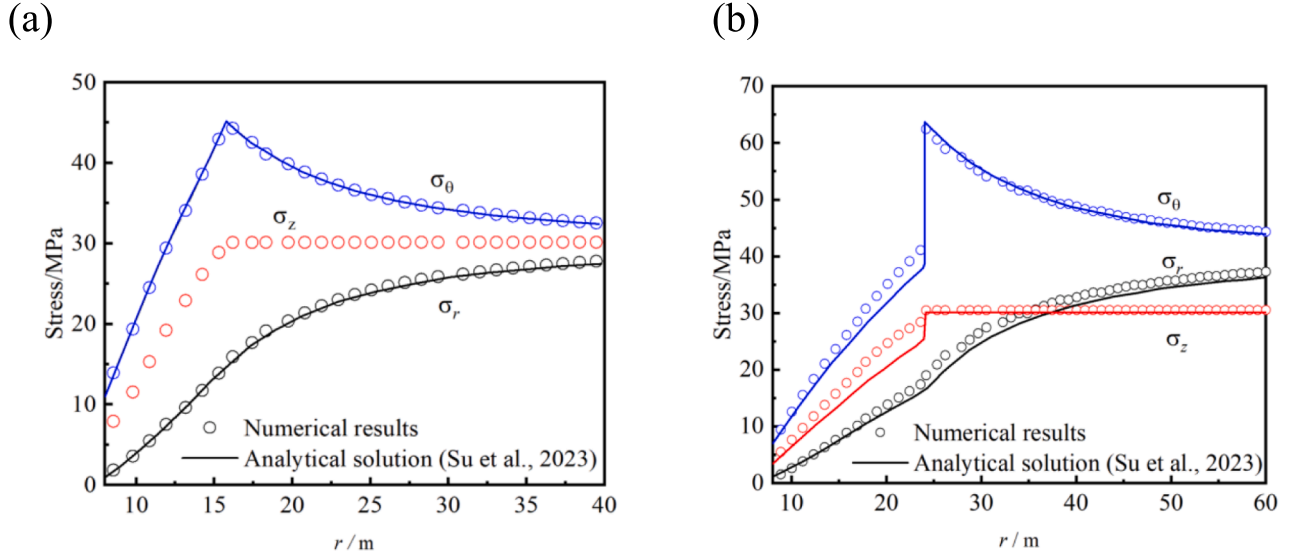
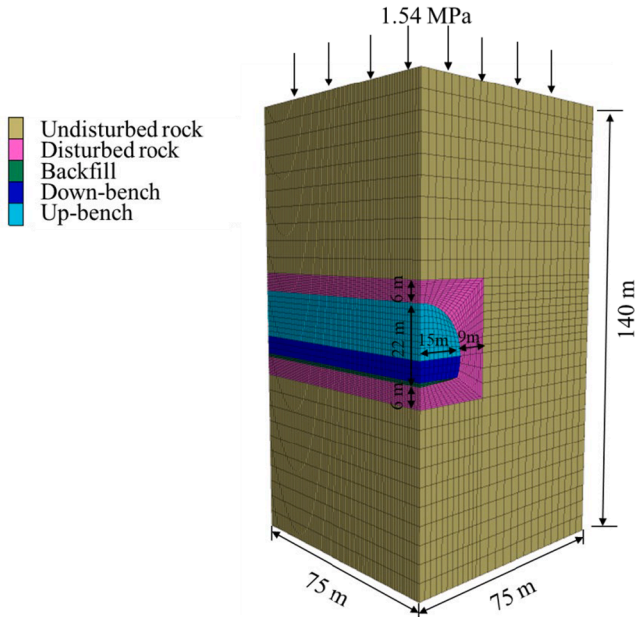
Fig. 13 shows the evolution of the roof settlement and horizontal convergence at the midpoint of the 0–2 m section. The results of the proposed GZZ-based model in this work agree well with the field measurements, including the deformation trend and magnitude. The slight difference between the results from the proposed model and the field measurements may be due to the difference in the rock behavior of elastic-perfectly plastic.

To further reveal the correctness of the proposed models, the computational results of several three-dimensional Hoek-Brown-based models (the modified GZZ-based model proposed by Zhu et al. (2017); the newly modified GZZ-based model proposed by Chen et al. (2021); and the GNUST-based model proposed by Wang et al. (2023)) and original Hoek-Brown-based model (Zhu et al., 2017) are also summarized in Fig. 13. The results calculated by the original Hoek-Brown-based model are both significantly larger than the field measurements and that of the three-dimensional Hoek-Brown-based models. The differences

Table 2

Model parameters for the circular tunnel. EP and EPB represent the elastic-perfectly plastic and elastic-brittle-plastic rock masses, respectively;

	E (GPa)	ν	m_i	D	GSI	σ_c (MPa)	GSI_r	p_0 (MPa)	σ_z (MPa)	p_i (MPa)
EP	2	0.27	6	0	50	40	50	30	30	1
EBP	9	0.3	6	0	65	50	27.2	40	30	1

**Fig. 11.** Comparison of the stress distributions from the analytical solution and numerical results adopting different constitutive models: (a) elastic-perfectly plastic constitutive model and (b) elastic-brittle-plastic constitutive model. (Analytical data from Su et al., 2023).**Fig. 12.** Numerical model of the highway rock tunnel.**Table 3**

Material parameters of the lining and backfill. (Zhu et al., 2017).

	Material	Density (kg/m ³)	E (GPa)	ν	Thickness (m)
Initial lining	Plain concrete	2200	21	0.25	0.2
Secondary lining	Reinforced concrete	2500	29.5	0.25	0.4
backfilling	Plain concrete	2400	26	0.25	1.2

between the results of the GZZ-based model and the Hoek-Brown-based model indicate that the intermediate principal stresses have a significant impact on the accuracy of the computational results. Nonetheless, the results of the proposed GZZ-based model in this work are almost coincidental with those of other three-dimensional Hoek-Brown-based models. The slight difference between the results from the proposed model and those from other three-dimensional Hoek-Brown-based models may be due to the difference between the finite element method and finite difference methods, the differences in the strength criterion and the difference in the flow rule.

Thus, the proposed GZZ-based constitutive model is validated through the strength and stress-strain curves in laboratory tests, the stress redistribution calculated from semi-analytical solutions, and the deformation in field and other three-dimensional Hoek-Brown-based models. It indicated that the proposed numerical implementation method provides an alternative solution to implement the GZZ criterion.

6. Conclusion

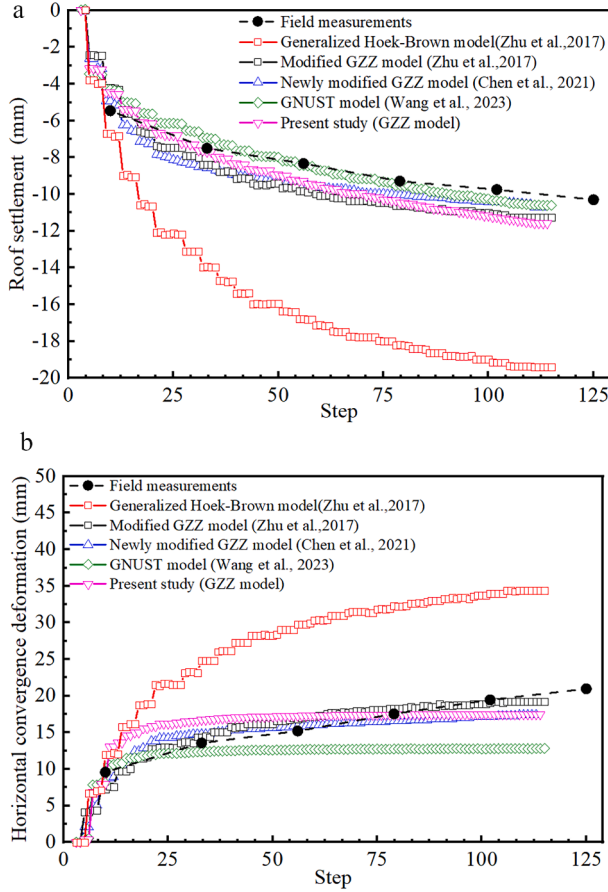
A novel numerical implementation method of a GZZ-based constitutive model, of which the yield surface is non-convexity and non-smoothness, is proposed in this work. To address the non-smoothness and non-convexity, the implemented method is performed in the $\sigma_m - \sqrt{J_2} - \theta_\sigma$ space. During the implementation of the nonlinear criterion in the $\sigma_m - \sqrt{J_2}$ space, the linearized nonlinear criterion is also adopted to avoid iterative calculations, greatly improving computational efficiency.

The proposed GZZ-based constitutive model can reproduce the volume expansion when adopting non-associated flow rule. Strain-softening can be realized using the table function in FLAC3D. The proposed GZZ-based model can also capture the strength and strain-strain curve obtained in laboratory tests well, the stress distribution induced by tunnelling, and the deformation measured in field engineering. In addition, the calculated deformation is consistent with that of the other

Table 4

Material parameters of the surrounding rock. (Zhu et al., 2017).

Rock type	Density (kg/m ³)	E (GPa)	ν	σ_c (MPa)	m_b	s	a	η
Disturbed	2350	3.24	0.3	13.2	1.835	0.0014	0.507	1.0
Undisturbed	2350	7.27	0.3	13.2	2.56	0.0027	0.507	1.0

**Fig. 13.** Comparison of the deformation from field measurements and numerical results for (a) roof settlement and (b) horizontal convergence.

three Hoek-Brown-based models.

Appendix A: Computing the plastic multiplier at stress-invariant space ($\sigma_m, \sqrt{J_2}, \theta_\sigma$)

As mentioned in Section 3.2, the nonlinear failure surface is continuously approximated by the linear tangent at the current stress. Once the stress state enters the plastic zone, the stress state should meet the following equation:

$$f(\sigma_m^P, \sqrt{J_2}^P, \theta_\sigma^P) = 0 \quad (\text{A.1})$$

$$f(\sigma_m^N, \sqrt{J_2}^N, \theta_\sigma^N) = f^*(\sigma_m^N, \sqrt{J_2}^N, \theta_\sigma^N) + C = 0 \quad (\text{A.2})$$

in which $\sigma_m^P, \sqrt{J_2}^P$ and θ_σ^P are the mean stress, shear stress, and Lode angle of the previous step, respectively. $\sigma_m^N, \sqrt{J_2}^N$ and θ_σ^N are the mean stress, shear stress, and Lode angle of the current step, respectively; $f()$ is the yield surface equation; $f^*()$ is the first-order term of the linear yield surface equation; and C is the constant term.

By substituting Eqs. (19) ~ (23), (36 ~ 37) and (A.1) into Eq. (A.2), the plastic multiplier can be obtained as follows:

$$\Delta\lambda = \frac{f(\sigma_m^I, \sqrt{J_2}^I, \theta_\sigma^I)}{G + \eta K q_\phi} \quad (\text{A.3})$$

The proposed numerical implementation method for the non-convexity and non-smoothness criterion is limited to the criterion of convexity and smoothness in $\sigma_m - \sqrt{J_2}$ at constant θ_σ . Further verification of the GZZ-based model is being carried out.

CRediT authorship contribution statement

Yingming Xiao: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Funding acquisition. **Manchao He:** Supervision. **Yafei Qiao:** Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing. **Wuqiang Cai:** Investigation. **Yujing Jiang:** Supervision. **Hehua Zhu:** Investigation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix B: Stress invariant state converted to the Cartesian stress state

A rescaling technique is adopted to convert the stress invariant state to the Cartesian stress state. The proportionality factor μ can be calculated as follows:

$$\mu = \frac{\sqrt{J_2^N}}{\sqrt{J_2}} \quad (\text{B.1})$$

The stress tensor components can be calculated as follows:

$$\sigma_{ij}^N = \sigma_m^N \delta_{ij} + \mu s_{ij}^I, j = 1, 2, 3 \quad (\text{B.2})$$

$$\delta_{ij} = \begin{cases} 1 & \text{when } i = j \\ 0 & \text{when } i \neq j \end{cases} \quad (\text{B.3})$$

in which s_{ij}^I is the stress deviation of the elastic guess.

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